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Learn2Unlock: Human–AI Synergy for Converting Children's Screen Time into KSSR Practice

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Highlights: Parental-control applications restrict children's screen time but rarely teach through it. Learn2Unlock converts device access into a reward earned by answering questions aligned to the Malaysian KSSR curriculum. A generative model drafts the questions, yet authority remains human: every draft passes a deterministic server-side filter and then explicit educator approval before a child sees it, and the model can never score an attempt or award a minute of screen time. A working parent application, child application and persistent API span four KSSR subjects, demonstrating a synergy in which AI scales content creation while educators keep full pedagogical control.

Keywords: human–AI synergy; human-in-the-loop; parental mediation; KSSR; educational technology

Introduction

Malaysian primary-school children are among the heaviest mobile device users in the region, and the tools available to the adults around them respond almost entirely with restriction: timers, blocklists and bedtime cut-offs. Restriction is adversarial by design — it positions the adult as an obstacle, generates daily negotiation, and teaches the child nothing.

A small category of answer-to-unlock applications tries to use that moment productively, but three problems have kept them marginal. Their question banks are generic and rarely map to the national curriculum a Malaysian child is assessed on. Where adults can add their own questions, they must author and maintain them by hand, which is unrealistic for a working parent or a teacher with thirty pupils. And where AI generates the questions it does so unsupervised, so the adult cannot inspect, correct or veto what the child is shown.

Learn2Unlock addresses all three, and does so in a way that speaks directly to the theme of human–AI synergy. The system does not ask anyone to trust the model: it confines the model to drafting, filters its output deterministically, and requires a human decision before any generated question can reach a child. AI supplies scale; the educator supplies judgement; neither substitutes for the other.

Objectives

- To make device access an outcome earned through curriculum-aligned practice rather than granted or withheld arbitrarily, so that every unlock episode produces retrieval practice on a named KSSR learning standard.
- To reduce the question-authoring effort required of parents and teachers by an order of magnitude through AI drafting, measured as the ratio of questions approved to questions written by hand.
- To guarantee architecturally that the language model cannot score an attempt, alter a rule or award screen time, so that adoption does not depend on trusting model behaviour.
- To deliver a working cross-platform system covering KSSR Years 1 to 6 in Mathematics, Science, Bahasa Melayu and English, with every item traceable to a learning-standard code.
- To report accuracy by learning standard, so that practice data feeds back into what is taught next.

NALI Approach

a) Novelty

Existing educational applications place artificial intelligence either fully in control of the learning loop or entirely outside it. Learn2Unlock introduces a third position: the model holds drafting authority only, and that boundary is enforced in code rather than in policy. Generated questions enter with the status pending_review and stay invisible to the child until an adult approves them; approval is the only path into the active bank. Scoring, reward calculation and rule enforcement run in deterministic server-side paths the model cannot reach. The novelty is not the reward loop, which is well established, but the placement of the human gate and the fact that it cannot be bypassed.

b) Creativity

Three design decisions carry the creative load. The deterministic pre-filter rejects a draft before any human sees it if it contains blocked terms, if the correct answer is absent from the options, if options duplicate after normalisation, if the explanation does not contain the correct answer, or if the item already exists in the bank — so adult review time goes to pedagogical judgement rather than to catching mechanical errors. The system ships as two separate applications rather than one with a PIN-protected settings panel, so the adult keeps a genuinely separate control surface. And emergency contacts and school applications stay reachable regardless of the balance, so the reward mechanism can never stand between a child and help.

c) Innovativeness

The improvement over current practice is measurable in adult effort and in curricular precision. Model output is constrained by a strict JSON schema at the API boundary, so malformed generations fail at parse time rather than propagating into the application. Every question, whether seeded, hand-written or AI-drafted, carries a subject, topic, KSSR learning-standard code, difficulty and language tag, making practice traceable to what the child is taught and enabling reporting at the level of the individual standard. The provider credential is held server-side only and never enters either mobile bundle.

d) Applicability

The design maps onto the New Academia Learning Innovation model in three respects. It is learner-centred: the child controls when to attempt a challenge and gets immediate feedback with an explanation for every item. It is technology-enhanced without being technology-governed, since the adult remains the decision-maker at the only point where content enters the system. And it is evidence-generating, because every attempt is logged against a learning standard and returned as a report.

Replication is straightforward. The taxonomy of learning standards is a data layer rather than a structural assumption, so substituting a different national curriculum, a secondary syllabus or a university course outline adapts the system without changing its architecture, and one teacher can approve a bank once and release it to many families.

Innovation Description, Process and Implementation

Learn2Unlock is a TypeScript monorepo of three deployable units. The parent application provides sign-in, a five-step family setup, a dashboard, additional-time approvals, question-bank management, manual authoring, AI generation and review, reports and settings. The child application provides the restriction notice, the challenge flow, the result and earned-time reward, a request to the parent, and an allowlist for emergency and school applications. A Node.js API server over SQLite handles authentication, family data, the review workflow, attempt scoring and the reward ledger; a shared package holds validation schemas and the KSSR-aligned seed bank. Table 1 summarises the authority each component holds.

In the interaction loop, a child opens a restricted application, sees the restriction notice, and begins a challenge of n questions, where n is set by the parent. The server scores the answers, and if the score meets the parent's passing threshold it credits reward minutes to the child's balance, subject to a daily cap and a maximum attempt count the parent also defines.

Table 1: Learn2Unlock components and the authority each one holds.

Component	Technology	Authority it holds
Parent application	Expo / React Native (TypeScript)	Defines all rules; approves, edits or rejects every question before a child sees it
Child application	Expo / React Native (TypeScript)	Attempts challenges and requests additional time; holds no credential for the model
API server	Node.js, SQLite, Zod schemas	Scores attempts, credits reward minutes, enforces daily caps, runs the deterministic safety filter
Language model	Server-side call, strict JSON schema	Drafts candidate questions only; cannot score, reward or modify a rule

The generation pipeline, shown in Figure 1, is where the division of labour becomes concrete. An adult specifies school year, subject, topic, KSSR learning-standard code, language, difficulty, question type and count. The request is issued from the server, never from the child's device and never with a credential in the mobile bundle. The model returns schema-constrained JSON, the deterministic filter rejects unsafe, duplicated or inconsistent items, and survivors are stored as pending_review for the adult to approve, edit or discard. Nothing that fails either gate is ever rendered on a child's screen.

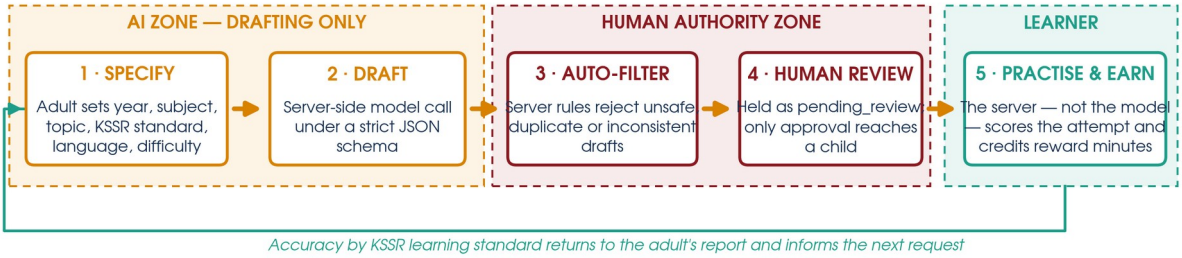


Figure 1: The Learn2Unlock human–AI pipeline. Authority passes from the model to the adult before any generated question can reach a child; scoring and reward stay outside the model entirely.

Effectiveness will be evaluated in a within-family, pre- and post-test pilot over eight weeks with [N] Malaysian families whose children are in KSSR Years 3 to 6, recruited through [recruitment channel]. Approval will be sought from [Institutional Research Ethics Committee]; consent and assent will be obtained and attempt data de-identified before analysis.

Key Findings

The current build is complete and operational end to end: both the parent flow (authentication, five-step onboarding, dashboard, requests, reports, settings) and the child flow (restriction, challenge, scored result, reward, request, emergency access) run against the persistent API. The seed bank covers four KSSR subjects — Mathematics, Science, Bahasa Melayu and English — every item tagged by topic, learning-standard code, difficulty and language. The generation endpoint runs with strict structured output and the full rejection filter, and the repository passes an automated API test suite and project-wide type checks.

Findings at this stage are therefore technical rather than behavioural, and no claim of learning gain is made until the pilot runs. The pilot will collect four classes of evidence: engagement, as attempts and accuracy trend by KSSR learning standard from the attempt log; conflict, as adult-reported frequency and intensity of screen-time disputes before and after; AI content quality, as the approval rate, edit rate and rejection reasons for AI drafts, which directly tests whether the pipeline reduces authoring burden without lowering standards; and usability and trust, using the System Usability Scale.

Impact

For students, minutes previously spent negotiating for a device become spaced retrieval practice on the exact standards their school assesses, with an explanation for every item. Empowerment is structural rather than rhetorical: the child decides when to attempt a challenge, and the reward is predictable and earned rather than granted at an adult's discretion.

For educators and parents, the innovation removes the authoring bottleneck that has made question-based approaches impractical. An adult who cannot supervise homework in real time can still shape what a child practises, in minutes of review rather than hours of tutoring, and one teacher's approved bank can serve many families at no additional cost. The wider contribution is a demonstration that generative AI can enter children's educational technology without asking anyone to trust an unsupervised model.

Recommendations for Classroom Practice

Teachers can adopt the system as a homework-practice channel: generate a bank against the learning standards for the current topic, review the drafts once, and release the approved set to the class. Batch review is far faster than authoring, and because the filter has already removed mechanically unsound items, attention can go to phrasing, difficulty and curricular fit.

Three practices are worth observing. Generate against one learning standard at a time, because the accuracy report is only as useful as the granularity of the tagging. Keep a mixture of difficulty levels in each release, so the challenge stays attainable for the weakest learner while stretching the strongest. And treat the rejection log as diagnostic: many rejected drafts usually indicate a standard phrased too broadly to generate against. Taxonomy, language and reward parameters are configuration, not code.

Other relevant information (if any)

The immediate market is Malaysian households with primary-school children, reachable through schools, PIBG networks and tuition centres, with a freemium tier, a paid family tier covering generation volume and reporting, and a school licence extending the review workflow to teachers. Server-side model calls keep unit economics controllable and credentials off user devices. No intellectual property has been filed to date; the software is an original work by the authors. [State any prior exhibition, award or IP filing here.]

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